Journal of Control, Automation and Electrical Systems (2019) 30:194–213
https://doi.org/10.1007/s40313-019-00439-0

Motion Analysis and Control of Three-Wheeled Omnidirectional Mobile Robot

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Received: 15 August 2018 / Revised: 9 November 2018 / Accepted: 2 January 2019 / Published online: 10 January 2019

Abstract

Omnidirectional mobile robots are holonomic vehicles that can perform translational and rotational motions independently and simultaneously. The paper provides a detailed mathematical analysis of the motion of a three-wheeled omnidirectional mobile robot leading to the kinematics of the robot. The motion of the robot can be divided into three types, pure rotation, linear motion and rotation around a point of a nonzero radius. The paper also addresses the problem of trajectory tracking, where the robot has to track the desired trajectory while tracking the desired orientation; to do so; a fuzzy controller has been designed. A comparison made between the proposed controller and another from the literature showed that the fuzzy controller with a minimal number of fuzzy rules (only four rules) is more efficient and more accurate. Furthermore, the paper proposes a simple approach to solve the kinematic saturation problem, namely that the control outputs must be within the range of the admissible control. A simulation platform was carried out using MATLAB to demonstrate the effectiveness of the proposed approach.

Keywords Three-wheeled omnidirectional mobile robot · Autonomous navigation · Trajectory tracking · Kinematic saturation · Fuzzy logic

1 Introduction

Nowadays, mobile robots are attracting more attention, and they are applicable in different environments such as ground applications (Ramalho et al. 2018), aerial applications (Viana et al. 2017; Silva et al. 2018) and underwater applications (Pezeshki et al. 2016). Wheeled mobile robots can be classified into *holonomic* or *non-holonomic* robots based on their mobility. *Non-holonomic* robots include the bicycle-type robot (Aleshin et al. 2017), differential drives (Al-Mayyahi et al. 2016), tricycle (Taniguchi, and Sugeno 2017) and car-like robot (de Lima and Pereira 2013; Wang et al. 2018a). A *Holonomic* robot is defined as a system with the same numbers of actuation and degrees of freedom. Omnidirectional mobile robots are *holonomic,* i.e., they can perform translational and rotational motions independently and simultaneously. The ability to move in any direction, regardless of the orientation of the vehicle, makes it an attractive option in different environments. So, they are very useful in many applications such as personal assistance (Moreno et al. 2016), rehabilitation (Kundua et al. 2017), industrial applications (Qian et al. 2017), service robots (Bae et al. 2017), hobby and competition (Aribowo et al. 2017), etc.

After the invention of the omnidirectional wheel by Grabowiecki (1919), various omnidirectional wheel mechanisms have been proposed in the last years, such as ball castors (Townsend 1958), Mecanum wheel (Ilon 1975), ball wheel (West and Asada 1997), longitudinal orthogonal-wheel (Mourioux et al. 2006), MY wheel (Ye and Ma 2009), MY wheel-II (Ma et al. 2012) and ACROBAT drive system (Inoue et al. 2013). Basically, this class of wheel mechanisms provides an active traction force in the ordinary direction, while it can be passively moving in the direction perpendicular to the active direction due to free rolling mechanisms.

Many research papers have studied the kinematic and dynamic modeling of the three-wheeled omnidirectional mobile robot (TWOMR) (Watanabe and Shiraishi 1998; Bi and Wang 2010; Ren and Ma 2015; Oliveira et al. 2009), but

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a few of them have studied its motion with analysis. Ashmore and Barnes (2002) analyzed the motion of omni-drive systems, where the motion of a simple two-wheeled omni-wheel system was described, and then, the velocity equation for an *n*-wheeled omni-drive robot has been derived; Ribeiro et al. (2004) described the motion of a TWOMR by considering the three types of motions: linear, angular and mixed motion, without establishing the kinematics of the robot. In Barua et al. (2015), after deriving the kinematics of the TWOMR, six different combinations of the three wheels velocities are tested by considering only the full speed of the wheel, one wheel is ON and two wheels are OFF, one wheel is OFF and two wheels are ON in the same direction, one wheel is OFF and two wheels are ON in opposite directions, three wheels are ON in the same direction, three wheels are ON in opposite directions and three wheels are ON where two wheels run in the same direction, while the third wheel runs in the opposite direction. Dong et al. (2016) have utilized a kinematics simulation analysis software to analyze two kinds of models T and equilateral triangle platform, and they showed just the results of the T-type model and summarized it by considering just four types of motions: self-rotation around the robot center, straight motion in the direction of 0°, oblique motion in the direction of 60° and lateral motion in the direction of 90°. Mohanraj et al. (2016) used an Angle Variable Chassis to study the effect of varying the angle between the two frontal wheels to the linear velocity of the TWOMR in both forward and backward directions; they have therefore studied a particular case which is the linear motion forward and backward. These works consider just common special cases of motion by considering motions in common directions or using special values such as the full velocity of the wheel, and miss the generality in the analysis of the TWOMR motion by taking all the possible values of the wheel velocity. In addition, they ignore the detailed analysis of the robot motion characteristics, for example, if we take the differential drive which has two driving wheels; by applying two velocity values to the wheels, one can easily find the linear and angular velocity of the robot, its heading, the Instantaneous Center of Curvature (ICC) and the instantaneous radius of rotation.

This paper is intended to complement the previous works by providing a detailed general analysis of the motion of the three-wheeled omnidirectional robot considering all possible values of wheel speed in both directions. The analysis also focuses on the relationships between the different forces and calculates from the three velocities applied to the wheels the motion characteristics: the linear velocity of the robot and its direction, the angular velocity of the robot, the linear velocities of wheels and their directions, the coordinates of the Instantaneous Center of Curvature (ICC), the instantaneous radius of rotation, and establishes the kinematics of the TWOMR based on a geometric approach. A simulation platform has been designed to clearly visualize the TWOMR motion and the different forces and to calculate the different parameters mentioned above. After establishing the kinematics of the TWOMR, the paper also addresses another problem, namely kinematic saturation, i.e., the controller must respect the maximum velocity of wheels during the generation of control outputs, this problem arises because the linear velocity and the angular velocity of the TWOMR can be controlled separately; in the literature, existing approaches have adopted the approximation to solve this problem by excluding a part of the possible velocities; in Kalmár-Nagy et al. (2004) and Wu et al. (2006), an approximation was adopted considering the internal values of the cone included in the inverted cube formed by the inverse kinematics and any value located outside this cone is not taken into account, the cone formula can be simply calculated. The same method was used for a Four-wheeled Omnidirectional Robot (Purwin and D'Andrea 2006). The paper, therefore, proposes a simple solution to the kinematic saturation problem of motors without excluding any possible velocity. After choosing the values of the linear and angular velocities, we apply the inverse kinematics to obtain the values of the velocities of the wheels which will be compared to the maximum velocity of the wheels. Furthermore, the trajectory tracking control was also considered, in which a fuzzy controller was designed and compared to another approach from the literature (Wang et al. 2018b).

How to consider the max velocity of the wheels

This paper is organized as follows. After the introduction, Sect. 2 presents a mathematical analysis of the TWOMR motion. Section 3 proposes a simple solution to the problem of kinematic saturation. Section 4 presents the trajectory tracking problem. The simulation results are given in Sect. 5, followed by the conclusion and future work.

2 Mathematical Analysis of the Three-Wheeled Omnidirectional Robot Motion

2.1 Description of the Robot

The three-wheeled omnidirectional mobile robot TWOMR is a *holonomic* robot that has the ability to move simultaneously and independently in translation and rotation. The robot is equipped with three omni-wheels equally arranged at 120 degrees on the circumference of the robot (Fig. 1). Each omni-wheel is mounted directly to its motor shaft so that the motor and the wheel have the same rotational axle. By individual control of the speed of each motor, the robot is able to perform motions that robots on normal wheels cannot.

Figure 2 shows an example of an omni-wheel where the wheel is equipped with many rollers to enable it to move sideways perpendicular to the normal direction of rolling. An

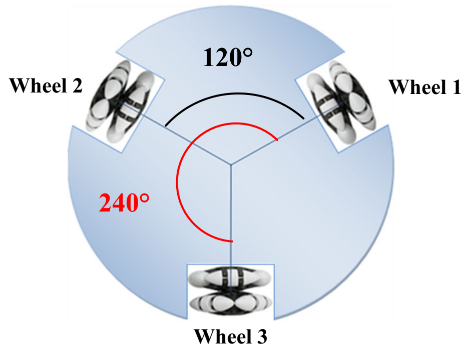


Fig. 1 The robot configuration

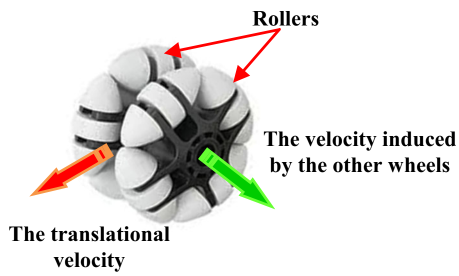


Fig. 2 The omni-wheel

omnidirectional drive system requires a minimum of three omni-wheels.

2.2 The Induced Velocity

Consider the omni-drive system of Fig. 3. Assume that the wheel 1 on the right is active with the translational velocity V_{w1}, while the wheel 2 and the wheel 3 on the left are inactive. Assuming there is no slippage, in this case, the wheel 2 and the wheel 3 will gain a velocity $V_{\text{in}1}$ perpendicular to their normal directions of rolling, outer of the wheel 2 and inner of the wheel 3 due to their rollers. This velocity is called the *induced velocity.* Therefore, the system will rotate about a single point C which is the intersection between the two lines perpendicular to the velocity of each wheel V_{w1} and $V_{\text{in}1}$

The induced velocity can be easily calculated. From Fig. 3, in triangle OBC we have:

$$\frac{R_2}{\sin 60} = \frac{L}{\sin 30} \tag{1}$$

where L refers to the robot radius, and therefore,

$$R_2 = \frac{L \sin 60}{\sin 30} = \sqrt{3}L \tag{2}$$

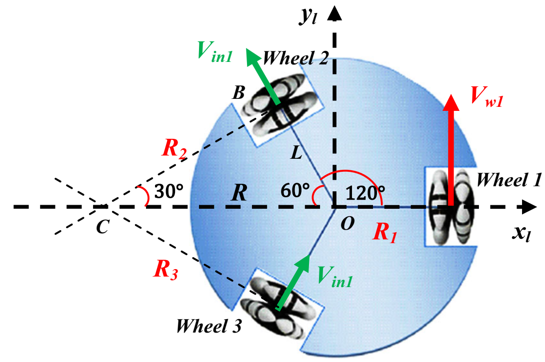


Fig. 3 Calculation of the induced velocity

Now, we can calculate the radius of rotation of the robot center R using *Pythagoras* theorem as follows:

$$R = \sqrt{R_2^2 + L^2} = 2L \tag{3}$$

Also,

$$R_1 = R + L = 3L \tag{4}$$

In this case depicted in Fig. 3, the robot turns around the point C, so the wheels of the robot turn with the same angular velocity:

$$\dot{\varphi} = \frac{V_{w1}}{R_1} = \frac{V_{\text{in}1}}{R_2} \tag{5}$$

Therefore,

$$V_{\text{in}1} = \frac{R_2}{R_1} = V_{w1} = \frac{V_1}{\sqrt{3}} \tag{6}$$

Then, the induced velocity $V_{\text{in}j}$ induced by applying the linear velocity V_{wj} to the wheel j is:

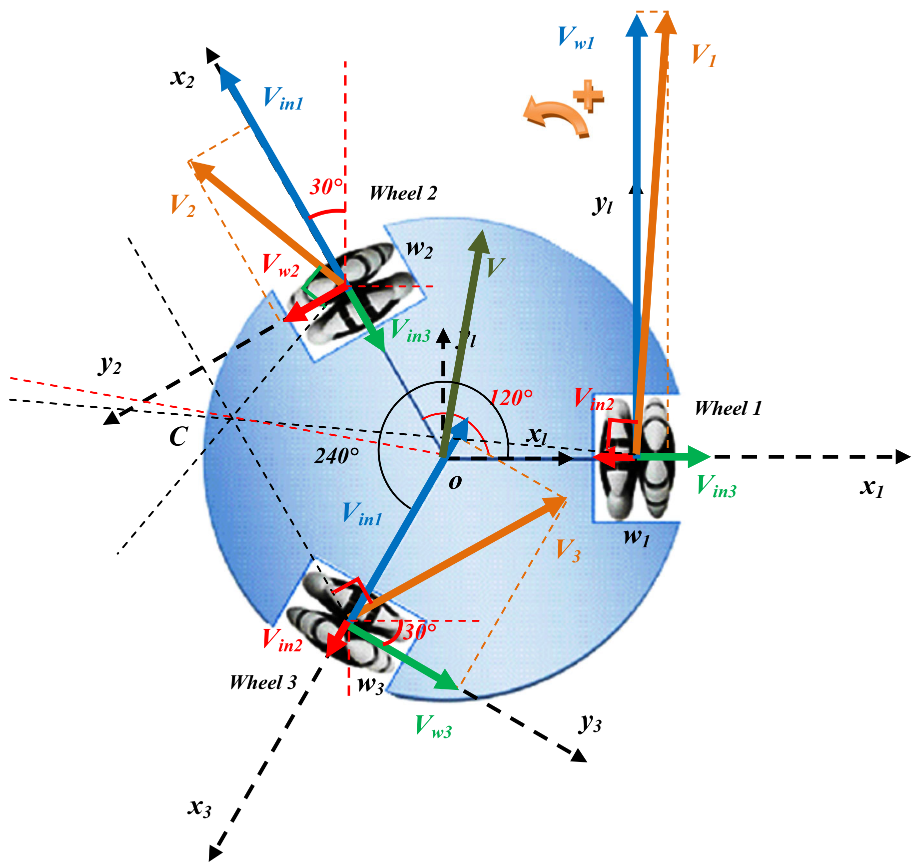
$$V_{\text{in}j} = \frac{V_{wj}}{\sqrt{3}} \tag{7}$$

This induced velocity is gained freely by the two rest wheels, and it is outer on the next wheel and inner on the precedent wheel. From Eq. (7), $V_{\text{in}j}$ has the same sign as V_{wj}.

2.3 Analysis of the Three-Wheeled Omnidirectional Robot Motion

Consider the general case illustrated in Fig. 4. Three velocities are applied to the three wheels V_{w1}, V_{w2} and V_{w3}. Consider the positive direction of rotation as the rotation in the counter-clockwise direction. We define a local frame (o, x_l, y_l) associated with the center of mass of the robot. The x_l-axis is parallel to the axis of rotation of the wheel 1. We attach

Fig. 4 Analysis of the different velocities on the three-wheeled omnidirectional robot



a local frame to each wheel, (w_1, x_1, y_1) to the wheel 1, (w_2, x_2, y_2) to the wheel 2 and (w_3, x_3, y_3) to the wheel 3. Each x_j-axis is parallel to the shaft of its wheel. The velocities V_{wj} applied to the wheels are taken with their algebraic values, i.e., if the velocity is in the positive direction, it is positive, and it is negative if it is in the negative direction. The transformations of velocities between the wheels' frames and the local frame of the robot can be given as follows:

$$\dot{X}_1^l = R_1^l \dot{X}_1 \tag{8}$$

$$\dot{X}_2^l = R_2^l \dot{X}_2 \tag{9}$$

$$\dot{X}_3^l = R_3^l \dot{X}_3 \tag{10}$$

where $\dot{X}_j$ ($j = 1{,}2$ and 3) is the coordinates of the velocity in the frame of the wheel j, while $\dot{X}_j^l$ is the coordinates of the velocity in the local frame of the robot. R_j^l, $j = 1$, 2 and 3 are the rotational matrices that transform the wheels' frames j to the local frame, and they can be written as:

$$R_1^l = \begin{bmatrix} \cos(0) & -\sin(0) \\ \sin(0) & \cos(0) \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \tag{11}$$

$$R_2^l = \begin{bmatrix} \cos(120) & -\sin(120) \\ \sin(120) & \cos(120) \end{bmatrix} = \begin{bmatrix} -\frac{1}{2} & -\frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & -\frac{1}{2} \end{bmatrix} \tag{12}$$

$$R_3^l = \begin{bmatrix} \cos(240) & -\sin(240) \\ \sin(240) & \cos(240) \end{bmatrix} = \begin{bmatrix} -\frac{1}{2} & \frac{\sqrt{3}}{2} \\ -\frac{\sqrt{3}}{2} & -\frac{1}{2} \end{bmatrix} \tag{13}$$

From Fig. 4, each applied wheel velocity V_{wj} is situated on the y_j direction of the wheel frame (w_j, x_j, y_j), while the induced velocities are situated on the x_j direction. Let V_j ($j = 1{,}2$ and 3) be the aggregate velocity applied to each wheel, so, its ordinate component is the wheel velocity V_{wj} applied to the wheel, and its abscissa component is the sum of the induced velocities. So, each aggregate velocity V_j can be expressed in its wheel frame (w_j, x_j, y_j) as follows:

$$\begin{cases} V_{1\_x1} = V_{\text{in}3} - V_{\text{in}2} = \dfrac{V_{w3}}{\sqrt{3}} - \dfrac{V_{w2}}{\sqrt{3}} \\ V_{1\_y1} = V_{w1} \end{cases} \tag{14}$$

$$\begin{cases} V_{2\_x2} = V_{\text{in}1} - V_{\text{in}3} = \dfrac{V_{w1}}{\sqrt{3}} - \dfrac{V_{w3}}{\sqrt{3}} \\ V_{2\_y2} = V_{w2} \end{cases} \tag{15}$$

$$\begin{cases} V_{3\_x3} = V_{\text{in}2} - V_{\text{in}1} = \dfrac{V_{w2}}{\sqrt{3}} - \dfrac{V_{w1}}{\sqrt{3}} \\ V_{3\_y3} = V_{w3} \end{cases} \tag{16}$$

Equations (14), (15) and (16) describe the aggregate translational velocity of each wheel in its wheel frame. To get these velocities in the local (robot) frame, we should transform them to the local frame as follows:

$$\begin{bmatrix} V_{1x} \\ V_{1y} \end{bmatrix} = R_1^l \begin{bmatrix} V_{1\_x1} \\ V_{1\_y1} \end{bmatrix} \tag{17}$$

$$\begin{bmatrix} V_{2x} \\ V_{2y} \end{bmatrix} = R_2^l \begin{bmatrix} V_{2\_x2} \\ V_{2\_y2} \end{bmatrix} \tag{18}$$

$$\begin{bmatrix} V_{3x} \\ V_{3y} \end{bmatrix} = R_3^l \begin{bmatrix} V_{3\_x3} \\ V_{3\_y3} \end{bmatrix} \tag{19}$$

By substituting (11–13) and (14–16) into (17, 18 and 19), respectively, we obtain:

$$\begin{cases} V_{1x} = -\frac{1}{\sqrt{3}} V_{w2} + \frac{1}{\sqrt{3}} V_{w3} \\ V_{1y} = V_{w1} \end{cases} \tag{20}$$

$$\begin{cases} V_{2x} = -\frac{1}{2\sqrt{3}} V_{w1} - \frac{\sqrt{3}}{2} V_{w2} + \frac{1}{2\sqrt{3}} V_{w3} \\ V_{2y} = \frac{1}{2} V_{w1} - \frac{1}{2} V_{w2} - \frac{1}{2} V_{w3} \end{cases} \tag{21}$$

$$\begin{cases} V_{3x} = \frac{1}{2\sqrt{3}} V_{w1} - \frac{1}{2\sqrt{3}} V_{w2} + \frac{\sqrt{3}}{2} V_{w3} \\ V_{3y} = \frac{1}{2} V_{w1} - \frac{1}{2} V_{w2} - \frac{1}{2} V_{w3} \end{cases} \tag{22}$$

Now, the resultant translational velocity of the robot V can be calculated by:

$$V = \frac{V_1 + V_2 + V_3}{3} \tag{23}$$

Then, their components are:

$$\begin{cases} V_x = \frac{V_{1x}+V_{2x}+V_{3x}}{3} \\ V_y = \frac{V_{1y}+V_{2y}+V_{3y}}{3} \end{cases} \tag{24}$$

Substituting Eqs. (20), (21) and (22) into (24) leads to:

$$\begin{cases} V_x = -\frac{\sqrt{3}}{3} V_{w2} + \frac{\sqrt{3}}{3} V_{w3} \\ V_y = \frac{2}{3} V_{w1} - \frac{1}{3} V_{w2} - \frac{1}{3} V_{w3} \end{cases} \tag{25}$$

To calculate the whole angular velocity $\dot{\varphi}$ of the robot, we use the contribution of each wheel. In the case depicted in Fig. 3, where the first wheel turns and has the translational velocity V_{w1}, while the two remaining wheels are looked, the robot turns around the point C, and therefore all points of the robot turn with the same angular velocity induced by V_{w1}. The contribution of the first wheel in the whole angular velocity of the robot can be given using Eqs. (4) and (5) by:

$$\dot{\varphi}_1 = \frac{V_{w1}}{3L} \tag{26}$$

Using the same way, we get the contribution of the second and the third wheels as follows:

$$\dot{\varphi}_2 = \frac{V_{w2}}{3L} \tag{27}$$

$$\dot{\varphi}_3 = \frac{V_{w3}}{3L} \tag{28}$$

Therefore,

$$\dot{\varphi} = \dot{\varphi}_1 + \dot{\varphi}_2 + \dot{\varphi}_3 = \frac{V_{w1} + V_{w2} + V_{w3}}{3L} \tag{29}$$

From Eqs. (25) and (29), we get:

$$\begin{cases} V_x = \dot{x} = -\frac{\sqrt{3}}{3} V_{w2} + \frac{\sqrt{3}}{3} V_{w3} \\ V_y = \dot{y} = \frac{2}{3} V_{w1} - \frac{1}{3} V_{w2} - \frac{1}{3} V_{w3} \\ \dot{\varphi} = \frac{1}{3L} V_{w1} + \frac{1}{3L} V_{w2} + \frac{1}{3L} V_{w3} \end{cases} \tag{30}$$

This equation is the *forward kinematics* of the TWOMR expressed in the local (robot) frame which can be written in matrix form as:

$$\begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{\varphi} \end{bmatrix} = \begin{bmatrix} 0 & -\frac{\sqrt{3}}{3} & \frac{\sqrt{3}}{3} \\ \frac{2}{3} & -\frac{1}{3} & -\frac{1}{3} \\ \frac{1}{3L} & \frac{1}{3L} & \frac{1}{3L} \end{bmatrix} \begin{bmatrix} V_{w1} \\ V_{w2} \\ V_{w3} \end{bmatrix} \tag{31}$$

The amplitude of the translational velocity can be given by:

$$V = \sqrt{V_x^2 + V_y^2} \tag{32}$$

And its direction by

$$\theta = \mathrm{atan2}(V_y, V_x) \tag{33}$$

where atan2 is the four-quadrant inverse tangent (arctangent) which can take its values in the interval [-π, π], while atan $\left(\frac{y}{x}\right)$ takes its values in the interval $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$.
The algebraic aggregate translational velocity of each wheel and its direction are given by:

$$V_j = \sqrt{V_{jx}^2 + V_{jy}^2}, \quad j = 1, 2 \text{ and } 3 \tag{34}$$

$$\theta_j = \mathrm{atan2}(V_{jy}, V_{jx}), \quad j = 1, 2 \text{ and } 3 \tag{35}$$

After obtaining the aggregate translation speed of each wheel, we can find the coordinates of the Instantaneous Center of Curvature (ICC), that is point C. by drawing the line perpendicular to each aggregate translational velocity V_j which passes through the center of the wheel w_j. The point

$C\ (C_x, C_y)$ is the intersection between these lines. First, the equation of each line will be found. The point $C\ (C_x, C_y)$ belongs to the line perpendicular to the vector V_j which verifies the relation:

$$\overrightarrow{w_j C}\begin{pmatrix} C_x - w_{jx} \\ C_y - w_{jy} \end{pmatrix} \perp \overrightarrow{V_j}\begin{pmatrix} V_{jx} \\ V_{jy} \end{pmatrix} \tag{36}$$

Therefore,

$$\left(C_x - w_{jx}\right)V_{jx} + \left(C_y - w_{jy}\right)V_{jy} = 0 \tag{37}$$

Equation (37) can be written as follows:

$$V_{jx}C_x + V_{jy}C_y - V_{jx}w_{jx} - V_{jy}w_{jy} = 0 \tag{38}$$

The coordinates of the wheels in the local frame are given as follows:

$$w_1 \begin{cases} w_{1x} = L \\ w_{1y} = 0 \end{cases} \tag{39}$$

$$w_2 \begin{cases} w_{2x} = L\cos 120 = -\frac{1}{2}L \\ w_{2y} = L\sin 120 = \frac{\sqrt{3}}{2}L \end{cases} \tag{40}$$

$$w_3 \begin{cases} w_{3x} = L\cos 240 = -\frac{1}{2}L \\ w_{3y} = L\sin 240 = -\frac{\sqrt{3}}{2}L \end{cases} \tag{41}$$

From (38), (39), (40) and (41), we get:

$$\begin{cases} V_{1x}C_x + V_{1y}C_y = V_{1x}L \\ V_{2x}C_x + V_{2y}C_y = -\frac{1}{2}V_{2x}L + \frac{\sqrt{3}}{2}V_{2y}L \\ V_{3x}C_x + V_{3y}C_y = -\frac{1}{2}V_{3x}L - \frac{\sqrt{3}}{2}V_{3y}L \end{cases} . \tag{42}$$

We have two unknown variables C_x and C_y, and then, we need two equations. We choose the second and the third equations to find the solution because of their symmetry that simplifies the solution.

$$\begin{cases} V_{2x}C_x + V_{2y}C_y = -\frac{1}{2}V_{2x}L + \frac{\sqrt{3}}{2}V_{2y}L \\ V_{3x}C_x + V_{3y}C_y = -\frac{1}{2}V_{3x}L - \frac{\sqrt{3}}{2}V_{3y}L \end{cases} \tag{43}$$

We calculate the determinant Δ,

$$\Delta = \begin{vmatrix} V_{2x} & V_{2y} \\ V_{3x} & V_{3y} \end{vmatrix} = V_{2x}V_{3y} - V_{3x}V_{2y} \tag{44}$$

- If $\Delta = 0$, there is no solution to (43). This corresponds to two parallel lines. Consequently, the velocity vectors are also parallel in the same direction and the robot moves in pure translation in the direction of these velocities perpendicular to these lines.
- If $\Delta \neq 0$, there is a solution that corresponds to the coordinates of the center of rotation C. Its coordinates are:

$$C_x = \frac{\begin{vmatrix} -\frac{1}{2}V_{2x}L + \frac{\sqrt{3}}{2}V_{2y}L & V_{2y} \\ -\frac{1}{2}V_{3x}L - \frac{\sqrt{3}}{2}V_{3y}L & V_{3y} \end{vmatrix}}{\Delta} \tag{45}$$

$$C_y = \frac{\begin{vmatrix} V_{2x} & -\frac{1}{2}V_{2x}L + \frac{\sqrt{3}}{2}V_{2y}L \\ V_{3x} & -\frac{1}{2}V_{3x}L - \frac{\sqrt{3}}{2}V_{3y}L \end{vmatrix}}{\Delta} \tag{46}$$

After substituting (20), (21), (22) and (44) into (45) and (46), and after simplification, we obtain:

$$C_x = L\frac{-2V_{w1} + V_{w2} + V_{w3}}{V_{w1} + V_{w2} + V_{w3}} \tag{47}$$

$$C_y = \sqrt{3}L\frac{-V_{w2} + V_{w3}}{V_{w1} + V_{w2} + V_{w3}} \tag{48}$$

The rotation radii correspond to the robot center, and the wheels can be easily calculated as follows: V_{w1}: Gesw. des 1. Rads

$$R = \sqrt{C_x^2 + C_y^2} = \frac{2L\sqrt{V_{w1}^2 + V_{w2}^2 + V_{w3}^2 - V_{w2}V_{w3} - V_{w1}V_{w2} - V_{w1}V_{w3}}}{V_{w1} + V_{w2} + V_{w3}} \tag{49}$$

L: Radius des Roboters

$$R_1 = \sqrt{\left(C_x - w_{1x}\right)^2 + \left(C_y - w_{1y}\right)^2} \tag{50}$$

$$R_2 = \sqrt{\left(C_x - w_{2x}\right)^2 + \left(C_y - w_{2y}\right)^2} \tag{51}$$

$$R_3 = \sqrt{\left(C_x - w_{3x}\right)^2 + \left(C_y - w_{3y}\right)^2} \tag{52}$$

From the kinematics equation, Eq. (30), we can calculate the translational and angular velocities of the robot:

$$\begin{cases} V = \sqrt{\dot{x}_l^2 + \dot{y}_l^2} \\ \dot{\varphi} = \frac{V_{w1}+V_{w2}+V_{w3}}{3L} \end{cases} \tag{53}$$

Beschreibung von V, $\dot{\varphi}$ und den Radius aus der Sicht des Zentrums des Roboters.

Then,

$$\begin{cases} V = \frac{1}{3}\sqrt{\left(-V_{w2} + V_{w3}\right)^2 + \left(2V_{w1} - V_{w2} - V_{w3}\right)^2} \\ \dot{\varphi} = \frac{V_{w1}+V_{w2}+V_{w3}}{3L} \end{cases} \tag{54}$$

From (54), we can summarize the robot motion as follows:

1. If $V_{w1} + V_{w2} + V_{w3} = 0$, then $\dot{\varphi} = 0$ and $V \neq 0$. This corresponds to an instantaneous pure translational motion.

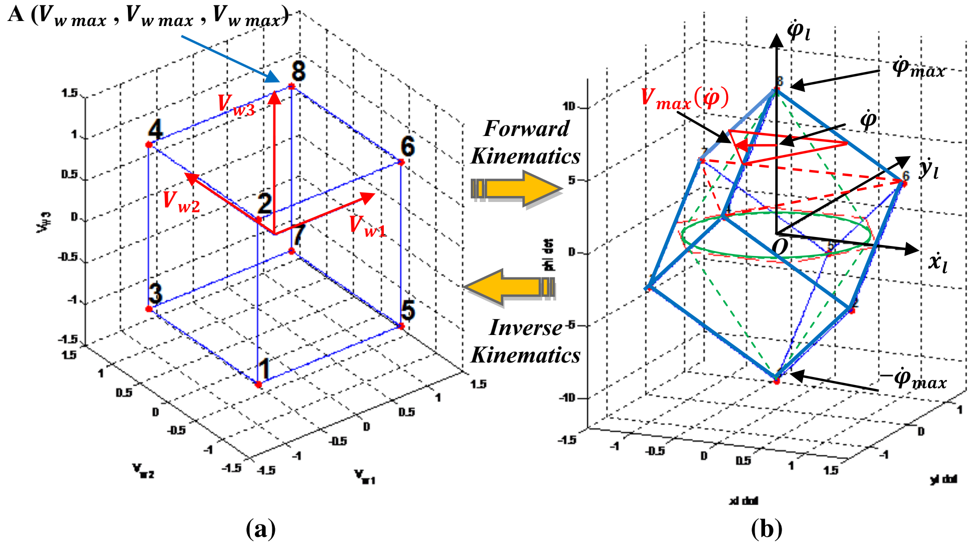


Fig. 5 Transformation between the world of the translational velocities of wheels and the local frame: **a** the world of the translational velocities of wheels and **b** the local frame

2. If $V_{w1} = V_{w2} = V_{w3}$, then $V = 0$ and $\dot{\varphi} \neq 0$. This corresponds to an instantaneous pure rotation.
3. If the two conditions mentioned above are not satisfied, the movement is on an instantaneous circular trajectory with the translational velocity V and the radius R.

The first case is equivalent to the case of $\Delta = 0$ where the robot moves in translational movement (i.e., the robot moves in a circle with the radius $R = \infty$). The second case is equivalent to the case where $\Delta \neq 0$, the robot moves in a circle with a radius $R = 0$ (pure rotation). The third case is equivalent to the case where $\Delta \neq 0$, and the robot moves in a circle with a radius $R \neq 0$.

3 The Kinematic Saturation

To design an efficient controller that allows the robot to respond correctly to its orders, the robot's angular and translational velocities must be within their ranges, depending on the actuators of the robot. Figure 5 shows the transformation between the frame of translational velocities of wheels and the local frame, where the possible translational velocities that can be taken with the wheels are enclosed in the blue cube (Fig. 5a).

Using forward kinematics Eq. (30), the image of the translational velocities of wheels in the local frame ($\dot{x}_l$, $\dot{y}_l$, $\dot{\varphi}_l$) appears as an inverted cube—parallelepiped (Fig. 5-b). The planned velocities V_{w1}, V_{w2} and V_{w3} (the admissible control) should be within the range of acceptable velocities, i.e., enclosed in the cube of Fig. 5a.

The omnidirectional robot can move simultaneously in rotation and translation. Thus, if it plans an angular velocity $\dot{\varphi} \in [-\dot{\varphi}\mathrm{max}, \dot{\varphi}\mathrm{max}]$, the planned translational velocity should not exceed the maximum translational velocity $V\mathrm{max}$ ($\dot{\varphi}$), this is illustrated in Fig. 5b).

Most approaches in the literature use the inverse kinematics to indirectly drive the wheels, where the kinematic saturation problem must be respected, i.e., the control outputs must be within the range of the admissible control. For the TWOMR, the admissible outputs must be enclosed in the inverted cube of Fig. 5b, and to get the allowable outputs we need a hard computation, because we cannot express an inverted cube using a simple formula, so we have to divide the inverted cube into several parts and do the computation which will be very hard. In the literature, this problem was solved by simplification. In Kalmár-Nagy et al. (2004) and Wu et al. (2006), an approximation was done based on the velocity cone (the dashed green cone in Fig. 5b), so that the velocity values used are enclosed in the green cone, and any velocity outside this cone is not taken into account. The same method was used for a Four Wheeled Omnidirectional Robot (Purwin and D'Andrea 2006).

In our paper, we propose a simple approach to obtain the admissible control outputs V_{wi}, where we consider all possible velocities in the inverted cube without excluding any possible velocity and without hard computation. Note that in the inverted cube (Fig. 5b), if we plan a velocity $\dot{\varphi} \in [-\dot{\varphi}\mathrm{max}, \dot{\varphi}\mathrm{max}]$, the maximum velocity $V\mathrm{max}(\dot{\varphi})$ belongs to a face of the inverted cube, i.e., if we apply the inverse kinematics to this maximum velocity, it belongs to a face in the velocity cube of Fig. 5a. So in our approach, we plan an angular velocity $\dot{\varphi} \in [-\dot{\varphi}\mathrm{max}, \dot{\varphi}\mathrm{max}]$, and then we plan a translational velocity V directed toward the target. And by using the projection on the x_l and y_l directions, we obtain the velocities $\dot{x}_l$, $\dot{y}_l$ and $\dot{\varphi}$. Then, by using the inverse kinematics we obtain the translational velocities of the wheels $\hat{V}_w = \left[\hat{V}_{w1}, \hat{V}_{w2}, \hat{V}_{w3}\right]^{\mathrm{T}}$. These velocities can be outside the allow-

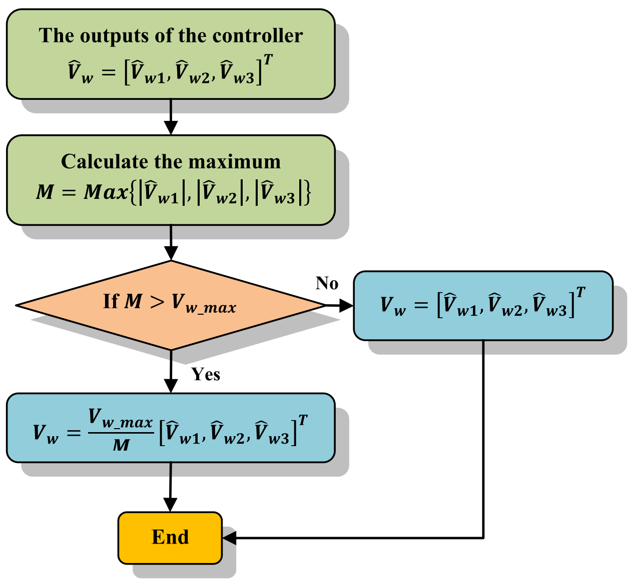


Fig. 6 The flowchart of the proposed solution to the saturation problem

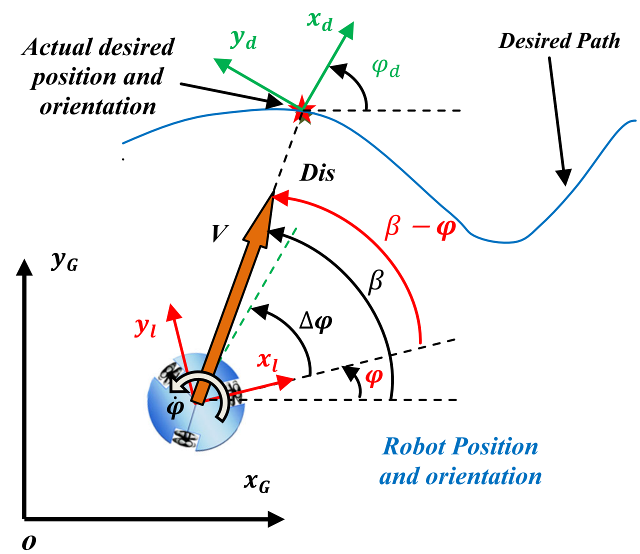


Fig. 7 The trajectory tracking problem

able values, so they must be limited by the maximum speed value.

Let $V_{w\,\max}$ be the maximum translational velocity of the wheels where the translational velocity V_{wi} should be in the range $-V_{w\,\max} \le V_{wi} \le V_{w\,\max}$. We calculate the maximum M of the velocities of the wheels with their absolute values and compare it to $V_{w\_\max}$.

$$M = \mathrm{Max}\left\{\left|\hat{V}_{w1}\right|, \left|\hat{V}_{w2}\right|, \left|\hat{V}_{w3}\right|\right\} \tag{55}$$

Then, the output is

$$V_w \begin{bmatrix} V_{w1} \\ V_{w2} \\ V_{w3} \end{bmatrix} = \begin{cases} \left[\hat{V}_{w1}, \hat{V}_{w2}, \hat{V}_{w3}\right]^{\mathrm{T}} & \text{if} \quad M \le V_{w\_}\max \\ \frac{V_{w\_}\max}{M}\left[\hat{V}_{w1}, \hat{V}_{w2}, \hat{V}_{w3}\right]^{\mathrm{T}} & \text{if} \quad M > V_{w\_}\max \end{cases} \tag{56}$$

The approach can be illustrated in the flowchart of Fig. 6.

4 Trajectory Tracking Control

The problem of trajectory tracking is depicted in Fig. 7. The robot has to track the desired trajectory and the desired orientation simultaneously. Since the TWOMR can move simultaneously in translation and rotation, the adopted strategy consists in applying a translational velocity V directed toward the target, in order to minimize the position error Dis (the distance between the robot and the desired position) and an angular velocity $\dot{\varphi}$ to minimize the orientation error $\Delta\varphi$ (the difference between the orientation of the robot and the desired orientation).

The distance Dis is given by:

$$Dis = \sqrt{(x_{\mathrm{d}} - x_G)^2 + (y_{\mathrm{d}} - y_G)^2} \tag{57}$$

The orientation error $\Delta\varphi$ is given by:

$$\Delta\varphi = \varphi_{\mathrm{d}} - \varphi \tag{58}$$

The overall control scheme of the trajectory tracking problem is depicted in Fig. 8. The inputs of the controller are Δx, Δy and $\Delta\varphi$ where:

$$\Delta x = x_{\mathrm{d}} - x_G \tag{59}$$

$$\Delta y = y_{\mathrm{d}} - y_G \tag{60}$$

The outputs of the controller are the three translational velocities of the wheels V_{w1}, V_{w2} and V_{w3}. The detailed structure of the controller is given in Fig. 9. The controller contains three modules; the first is the internal controller module which can be a PID, a neural, a fuzzy, etc. The internal controller has two inputs Dis and $\Delta\varphi$ and two outputs V and $\dot{\varphi}_l$.

The second module is the inverse kinematics module which has three inputs $\dot{x}_l$, $\dot{y}_l$ and $\dot{\varphi}_l$ where

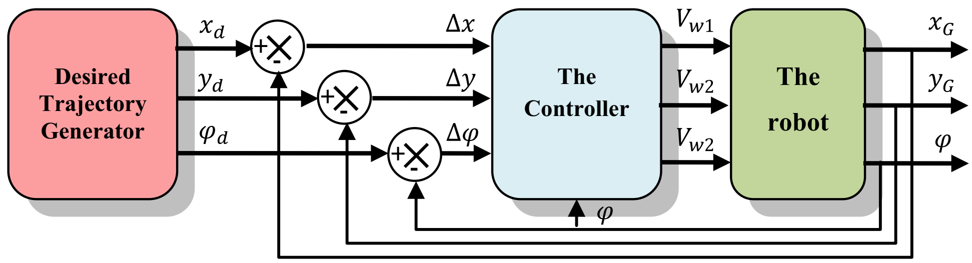
$$\dot{x}_l = V\cos(\beta) \tag{61}$$

$$\dot{y}_l = V\sin(\beta) \tag{62}$$

The bearing angle β is given by:

$$\beta = \mathrm{atan2}(\Delta y, \Delta x) \tag{63}$$

Fig. 8 The control scheme of the trajectory tracking problem



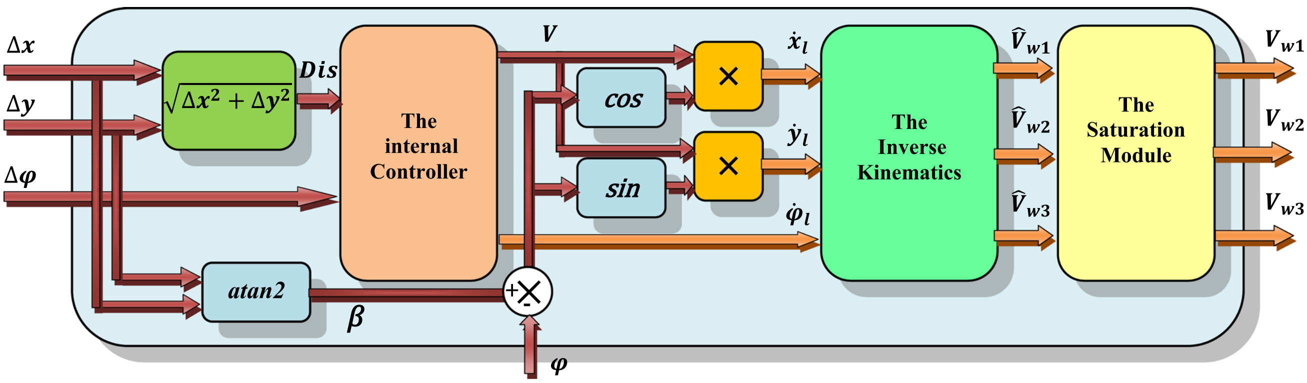


Fig. 9 The controller structure

where atan2(y, x) is the *four-quadrant inverse tangent* (arctangent) of y and x, and whose results belong to the closed interval $[-\pi, \pi]$, while atan $\left(\frac{y}{x}\right)$ takes its values in the interval $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$. φ is the heading angle of the robot and $\beta - \varphi$ is the relative bearing angle (in the local frame of the robot).

The outputs of the inverse kinematics module are the velocities of wheels $\hat{V}_{w1}$, $\hat{V}_{w2}$ and $\hat{V}_{w3}$. It is not guaranteed that these velocities do not exceed the maximum velocity of the wheels $V_{w\_\max}$; we must, therefore, limit them with the saturation module based on the flowchart of Fig. 6.

In the following, we propose a fuzzy controller.

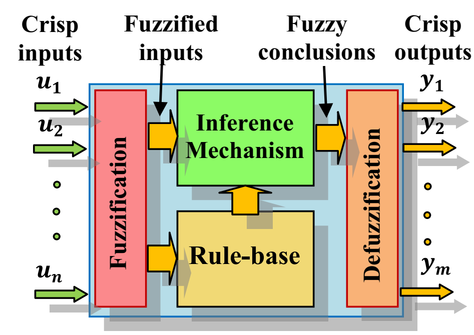


Fig. 10 General fuzzy controller

4.1 The Fuzzy Controller

The theory of fuzzy logic systems is inspired by the remarkable human capacity to reason with perception-based information. Rule-based fuzzy logic provides a formal methodology for linguistic rules resulting from reasoning and decision making with uncertain and imprecise information. A fuzzy controller is a static nonlinear mapping between its inputs and outputs (i.e., it is not a dynamic system). The inputs and outputs are "crisp," that is, they are real numbers, not fuzzy sets.

The fuzzy controller has four main components (Fig. 10):

- The *fuzzification* block: converts the crisp inputs to fuzzy sets,
- The *rule base*: holds the knowledge, in the form of a set of rules, of how best to control the system,
- The *inference mechanism*: The inference mechanism evaluates which control rules in the *rule base* are relevant at the current time to produce fuzzy conclusions and then decides what the input to the plant should be (e.g., the implied fuzzy sets), and
- The *defuzzification* block: converts the fuzzy conclusions reached by the inference mechanism into the crisp outputs, which are the inputs to the plant.

The fuzzy controller is given in Fig. 11, and it comprises two inputs: the distance between the robot and the desired position *Dis* and the difference between the orientation of the robot and the desired orientation $\Delta\varphi$; and two outputs, the translational velocity V and the angular velocity $\dot{\varphi}$.

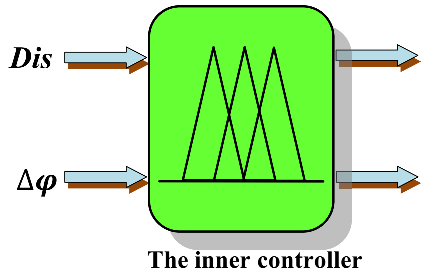


Fig. 11 The fuzzy controller

Table 1 Linguistic values of Dis

Linguistic term	Label
Zero	Z
Far	F

Table 2 Linguistic values of V

Linguistic term	Label
Zero	Z
Big	B

Table 3 Linguistic values of $\Delta\varphi$ and $\dot{\varphi}$

Linguistic term	Label
Zero	Z
Negative	N
Positive	P

4.2 The Fuzzification Procedure

Different types of membership functions are used. Triangular and trapeze membership functions are used for the fuzzy values of the two inputs Dis and $\Delta\varphi$ and the second output $\dot{\varphi}$. For the first output V, the Gaussian membership functions are used. The labels used to express the values of Dis are given in Table 1, those of the translational velocity V are given in Table 2, and those of $\Delta\varphi$ and $\dot{\varphi}$ are given in Table 3.

The membership functions for all the terms of the input and output variables in this controller are shown in Fig. 12.

4.3 The Rule Base Design

The rule base of the fuzzy controller contains four rules as shown in Table 4.

4.4 The Defuzzification Procedure

The "*centroid*" method is used for the defuzzification procedure. Centroid defuzzification returns the center of the area under the curve.

5 Simulation Results

The simulation was performed with MATLAB. This section begins with the simulation of the detailed analysis of the movements. The performance of the proposed approach was carried out using two tests, point stabilization and trajectory tracking, and compared with the results of the paper by Wang et al. (2018b). In Wang et al. (2018b), a model predictive control (MPC) algorithm was designed to achieve point stabilization and trajectory tracking of a three-wheeled omnidirectional robot. To make the comparison, the same conditions and characteristics as those used in Wang et al. (2018b) are used, the maximum velocity of the wheels is 2 m/s ($V_{w}\_max = 2\,\text{m/s}$). In addition, complex scenarios are performed at the end of this simulation.

5.1 Simulation of the Robot with Motion Analysis

The motion of the robot is simulated for the three general cases. First, we simulate the robot in the pure rotation around its center of mass, then, in the pure translational motion, finally, the rotation around an arbitrary point C of radius R. For a clearer graphical representation, the velocities are represented with different colors as follows: the translational velocities of the wheels V_{wj} are represented in red color, the induced velocity $V_{\text{in}j}$ in blue color, the aggregate translational velocity V_j in black color, the translational velocity of the robot center V in yellow color, and the center of rotation C is represented with a red point-like circle.

5.1.1 Pure Rotation

The pure rotation of the robot around its center of mass can be achieved by driving all the wheels in the same direction and at the same speed ($V_{w1} = V_{w2} = V_{w3}$). Figure 13 shows an example where $V_{w1} = V_{w2} = V_{w3} = 0.5$ m/s.

5.1.2 Linear Movement

Linear movement (Fig. 14) occurs when the algebraic sum of the velocities of wheels is zero ($V_{w1} + V_{w2} + V_{w3} = 0$). In this scenario, $V_1 = 1$, $V_2 = -0.7$ and $V_3 = -0.3$.

5.1.3 Rotation Around a Point C with a Radius R

When the two conditions above are not verified, the robot moves in rotational motion around a point C different from the robot center with a radius $R \neq 0$. This case is depicted in Fig. 15.

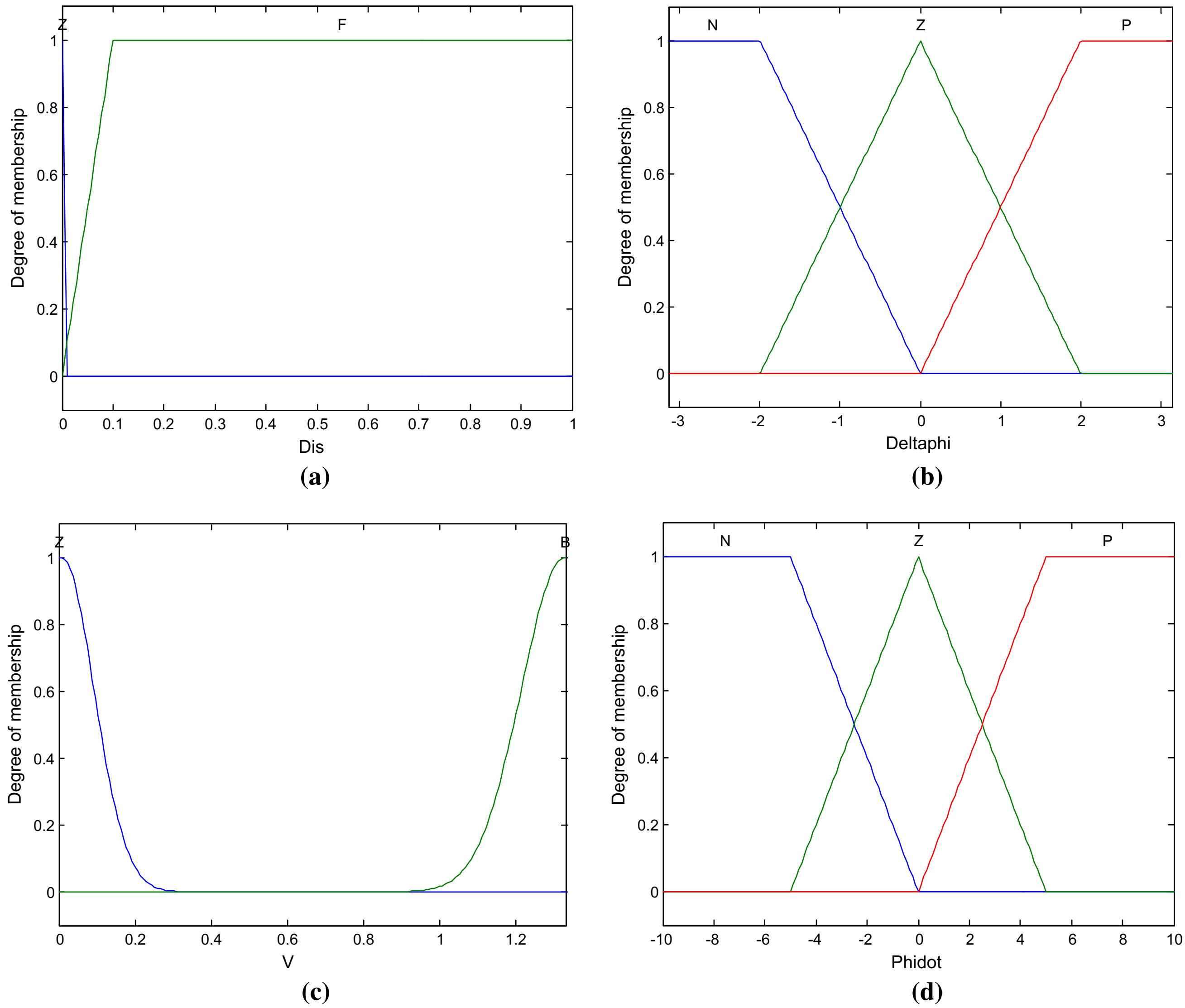


Fig. 12 The membership functions of: **a** Dis, **b** $\Delta\varphi$, **c** V and **d** $\dot{\varphi}$

Table 4 The rule base

N	Dis	$\Delta\varphi$	V	$\dot{\varphi}$
1	Z	N	Z	Z
2	F	N	B	N
3	F	Z	B	Z
4	F	P	B	P

5.2 Point Stabilization

Point-to-point stabilization consists of having the robot to reach a steady target located at a distance to the robot and staying at this location, and this is similar to the controller step response. The results are compared with those of the article by Wang et al. (2018b). The same conditions and scenario characteristics are used. The robot initially is located at the origin with the orientation (0 rad), and the desired target is located at the point (3 m, 2 m), while the desired orientation is $\pi/3$ rad.

The results of both the proposed approach and the approach by Wang et al. (2018b) are illustrated in Fig. 16. Compared to the approach proposed by Wang et al. (2018b), we can see that the robot with the proposed approach (Fig. 16a) can reach the target in the fastest way by following the shortest path that is the straight line between the origin (0 m, 0 m) and the target (3 m, 2 m), whereas in the approach

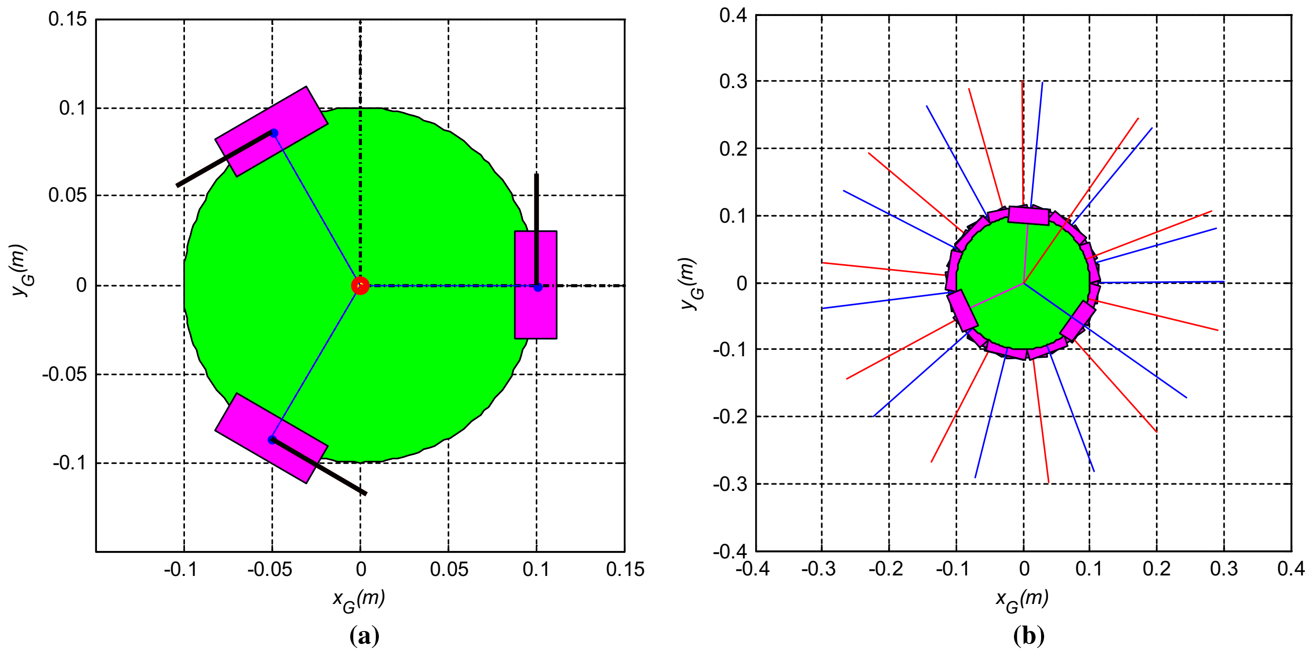


Fig. 13 Pure rotation ($V_{w1} = V_{w2} = V_{w3} = 0.5$ m/s): **a** analysis of the different velocities and **b** the rotational motion in the global frame

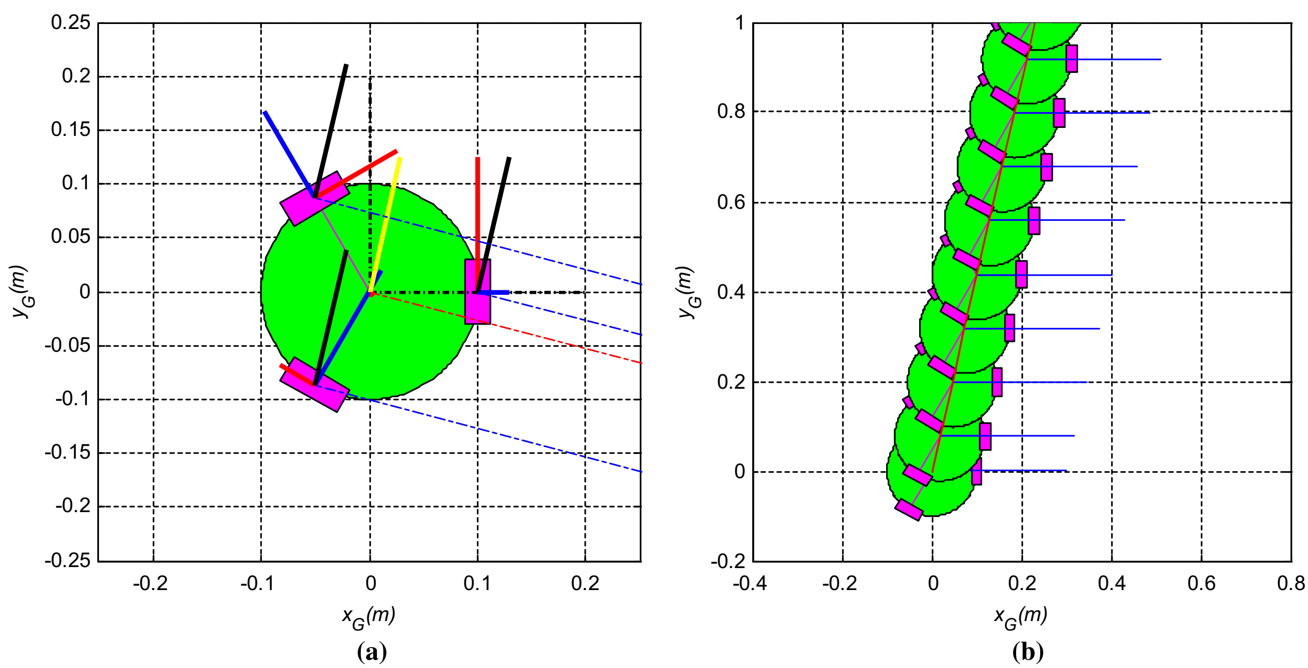


Fig. 14 Pure translational motion, where $V_{w1} + V_{w2} + V_{w3} = 0$, in this example ($V_{w1} = 1$, $V_{w2} = -0.7$, $V_{w3} = -0.3$): **a** analysis of the different velocities and **b** the rotational motion in the global frame

of Wang et al. (2018b) the robot can reach the target but with deviation from the desired path (Fig. 16b).

The x and y coordinates of the trajectory are given in Fig. 17. The initial pose has been set to be (0 m, 0 m, 0 rad) and the desired pose has been set to be (3 m, 2 m, $\pi/3$). Figure 17 shows that the response time of the proposed approach (Fig. 17a) is about 2.23 s which is less than the response time of Wang et al. (2018b) which is about 5 s (Fig. 17b).

The orientation is given in Fig. 18; it shows that the proposed approach is faster and more accurate than that of Wang et al. (2018b), and its curve has no oscillations.

The velocities of wheels employed in this scenario are given in Fig. 19. In the left figure (Fig. 19a), the two

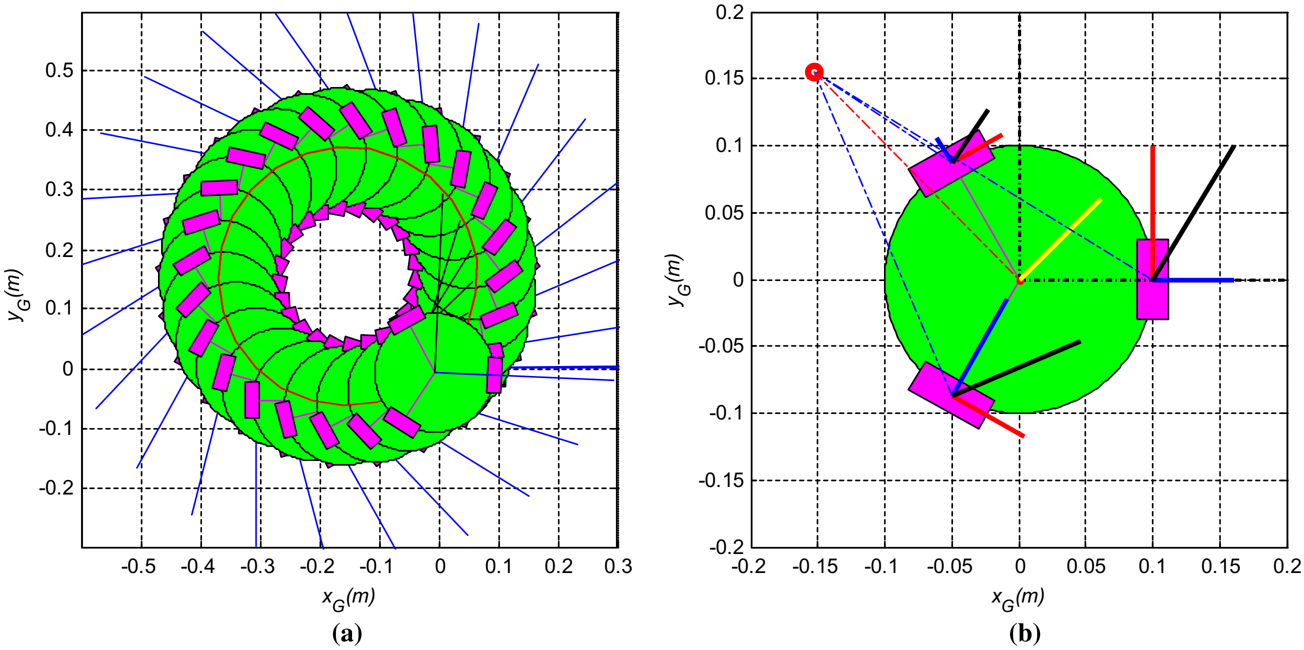


Fig. 15 Rotation around a point C with a radius R: **a** analysis of the different velocities and **b** the rotational motion in the global frame

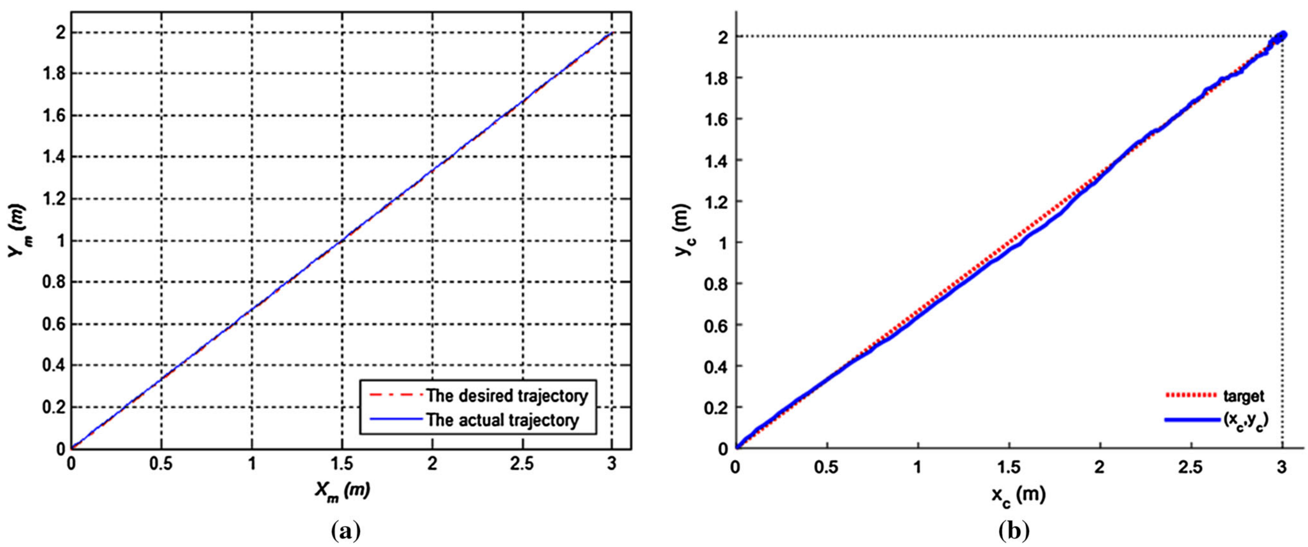


Fig. 16 Point stabilization: **a** the proposed approach and **b** the approach of Wang et al. (2018b)

green lines represent the range of the wheel velocity $V_{wi} \in [-2\,\text{m/s}, 2\,\text{m/s}]$. At the beginning, the three wheels turn approximately at the maximum velocity (2 m/s) which corresponds to the rotation in the positive direction (counter-clockwise direction) to track the desired orientation ($\pi/3$). Then, the robot moves in a straight line toward the target. If we calculate the sum of the velocities of the three wheels we get zero, which corresponds to the linear motion. After 2.23 s, the robot reaches the target and stays there the rest of the time. In Wang et al. (2018b), the time required to reach the target is being approximately 5 s, so the proposed approach is faster than that of Wang et al. (2018b).

The tracking error is given in Fig. 20. The results show that the maximum tracking error of the proposed approach is less than 1 mm, whereas that of Wang et al. (2018b) is about 4 cm, which shows that the proposed approach is more accurate than that of Wang et al. (2018b).

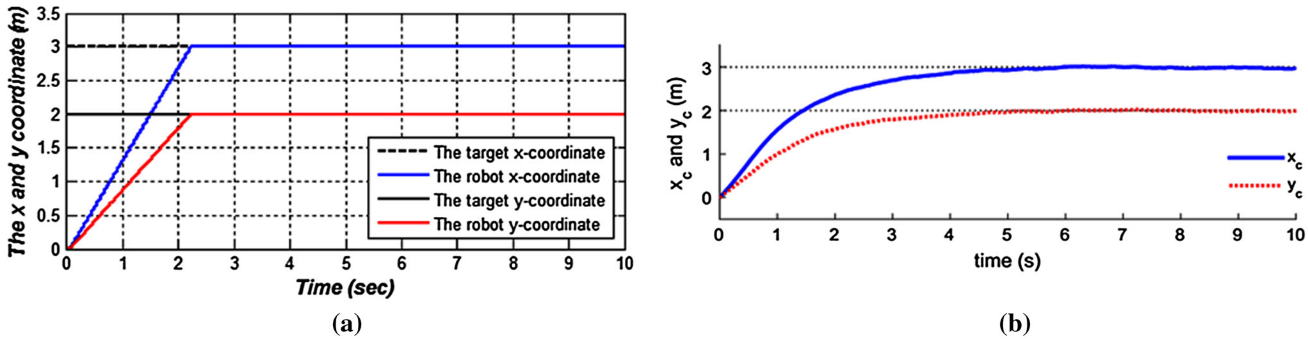


Fig. 17 The *x* and *y* coordinates of the trajectory of point stabilization: **a** the proposed approach and **b** the approach of Wang et al. (2018b)

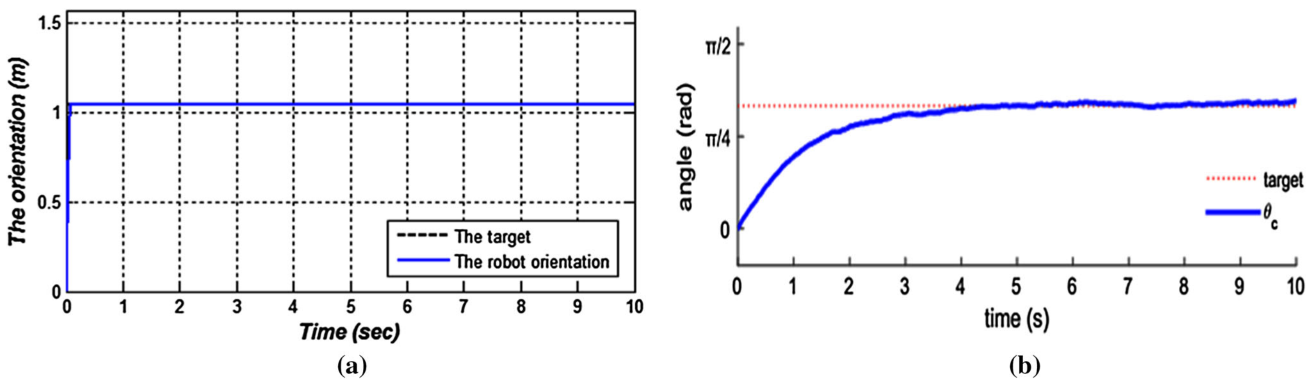


Fig. 18 Comparison between the robot orientation and the desired orientation: **a** the proposed approach and **b** the approach of Wang et al. (2018b)

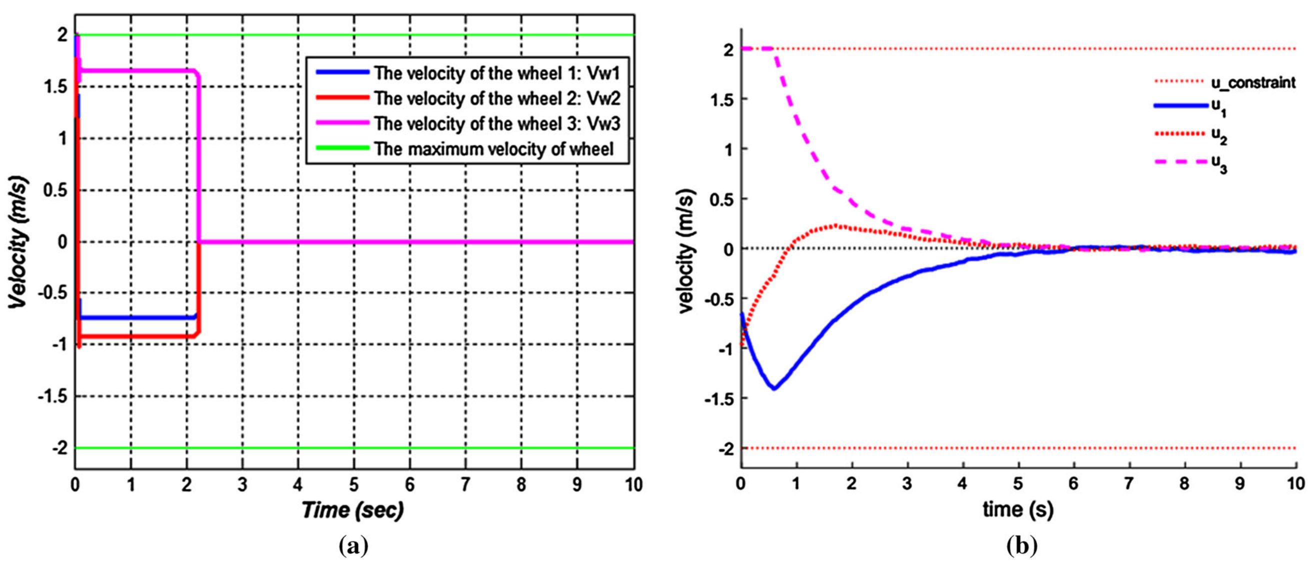


Fig. 19 The velocity of wheels: **a** the proposed approach and **b** the approach of Wang et al. (2018b)

5.3 Trajectory Tracking

In this test, the robot has to track a pulse trajectory. The same conditions are used as in the work of Wang et al. (2018b). The pulse trajectory contains six nodes; their poses are given in Table 5. The comparisons of trajectory tracking and orientation between the two approaches are illustrated in Figs. 21 and 22, respectively. As shown in Fig. 21a, the robot with the proposed approach can reach each node in the shortest path, namely the straight line between two successive nodes without deviation from the desired path. On the other hand, in the approach of Wang et al. (2018b), the robot deviates from the

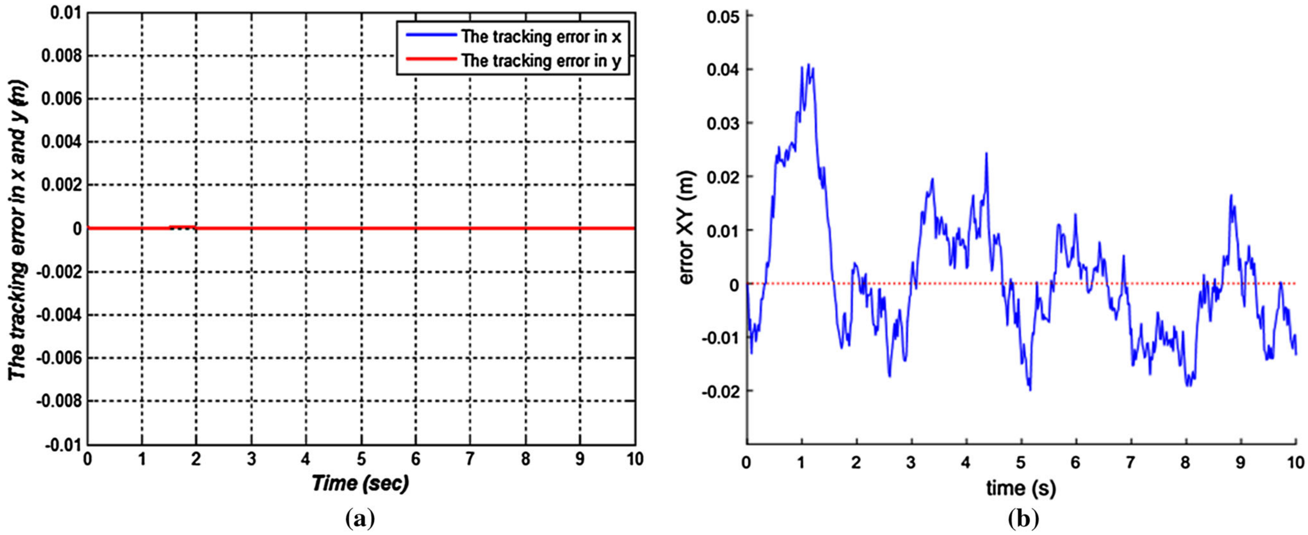


Fig. 20 The tracking error of wheels: **a** the proposed approach and **b** the approach of Wang et al. (2018b)

Table 5 The node parameters of the trajectory tracking

The node	x-coordinate (m)	y-coordinate (m)	Orientation
1	0	1	0
2	2	1	0
3	2	3	$\pi/2$
4	4	3	0
5	4	1	$-\pi/2$
6	6	1	0

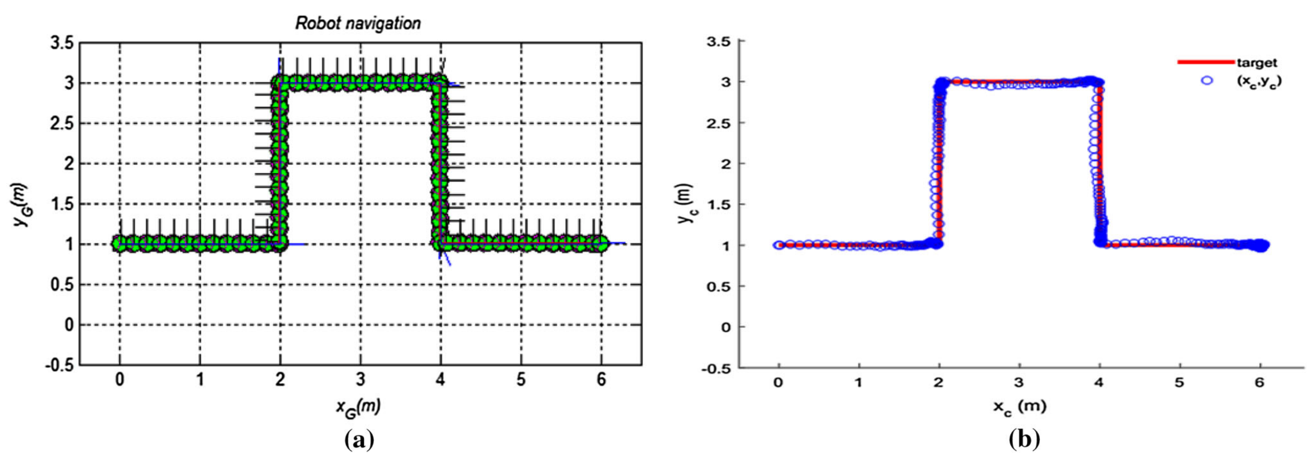


Fig. 21 Tracking a pulse trajectory: **a** the proposed approach and **b** the approach of Wang et al. (2018b)

desired path. Also, the robot with the proposed approach can reach the desired orientation faster than the approach of Wang et al. (2018b). In Fig. 22a, the desired orientation between two successive nodes can be reached in 0.13 s, which means that the robot maintains the desired orientation about 90% of the time spent from one node to another. In Fig. 22b, the robot consumes almost all the time spent between two successive nodes to reach the desired orientation and maintains it only for a short time.

The linear velocity and the angular velocity are shown in Figs. 23 and 24, respectively. We can see that the robot has completed the trajectory in a short time, about 6.37 s, while in the approach of Wang et al. (2018b), the time needed to complete the course is about 30 s. In addition, the robot with the proposed approach can quickly achieve the desired

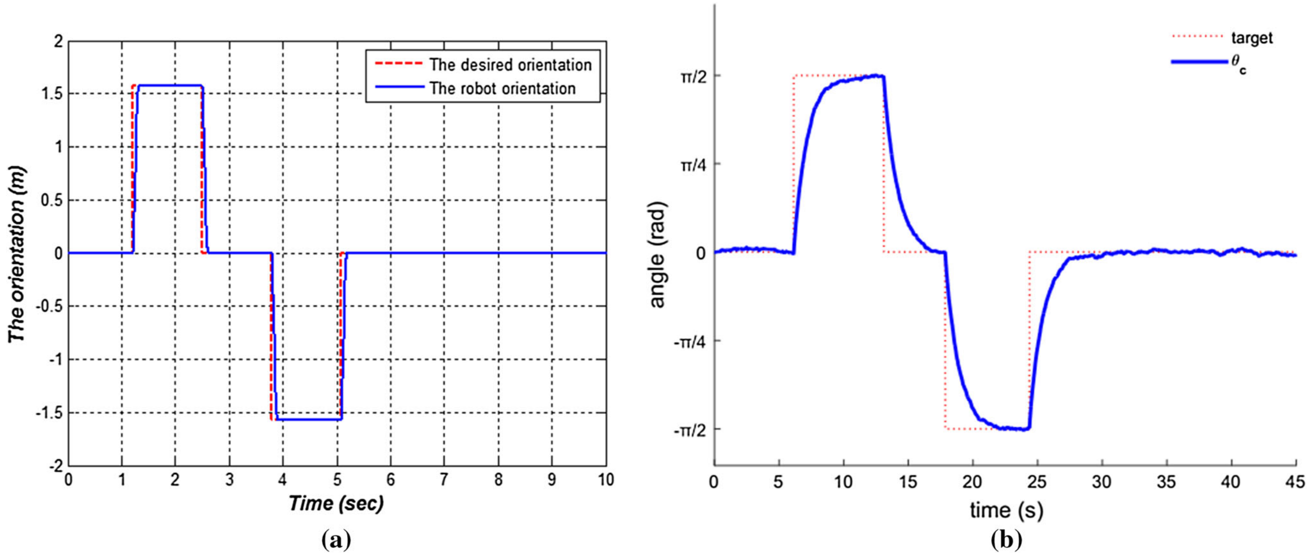


Fig. 22 The orientation of tracking a pulse trajectory: **a** the proposed approach and **b** the approach of Wang et al. (2018b)

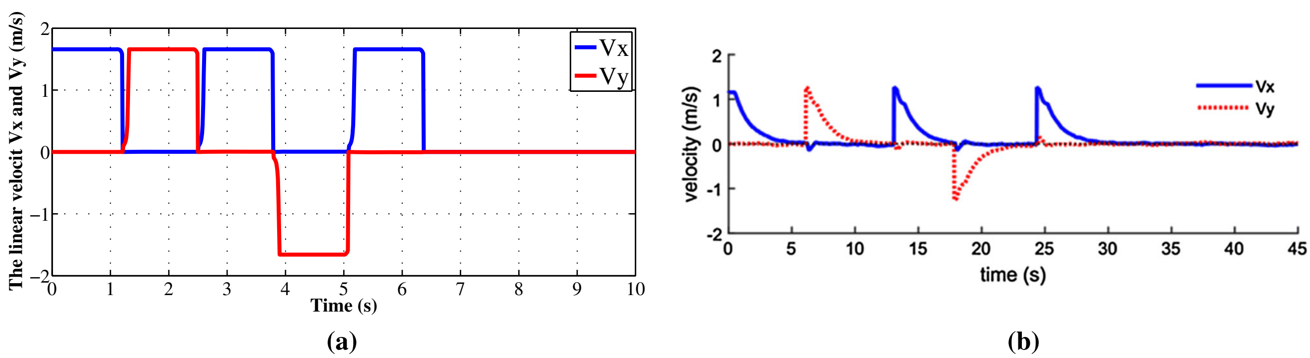


Fig. 23 The robot linear velocity of tracking a pulse trajectory: **a** the proposed approach and **b** the approach of Wang et al. (2018b)

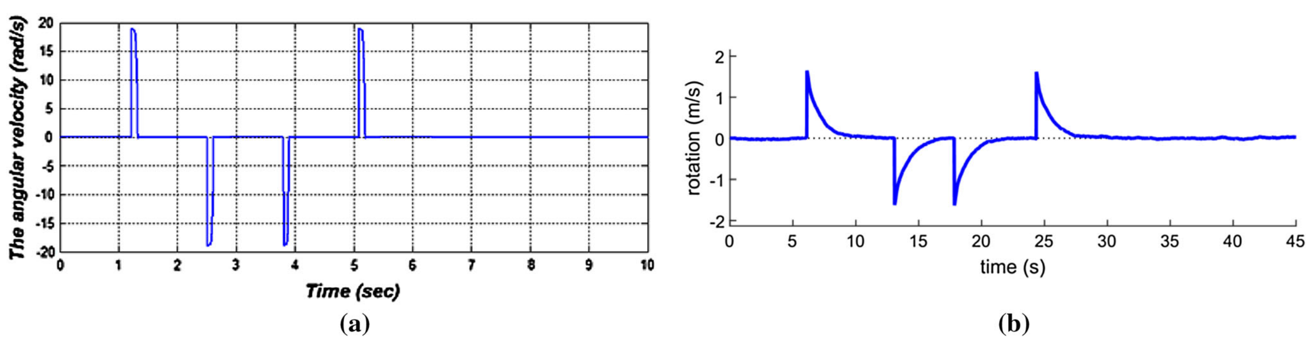


Fig. 24 The robot angular velocity of tracking a pulse trajectory: **a** the proposed approach and **b** the approach of Wang et al. (2018b)

orientation compared to that of Wang et al. (2018b), which takes longer.

The trajectory tracking error is depicted in Fig. 25. The maximal tracking error of the proposed approach is less than 1 cm (Fig. 25a), while that of Wang et al. (2018b) is about 8 cm (Fig. 25b). The two previous tests, point stabilization and trajectory tracking, demonstrate the effectiveness of the proposed approach, compared to the approach used in Wang et al. (2018b), and it is faster and more accurate.

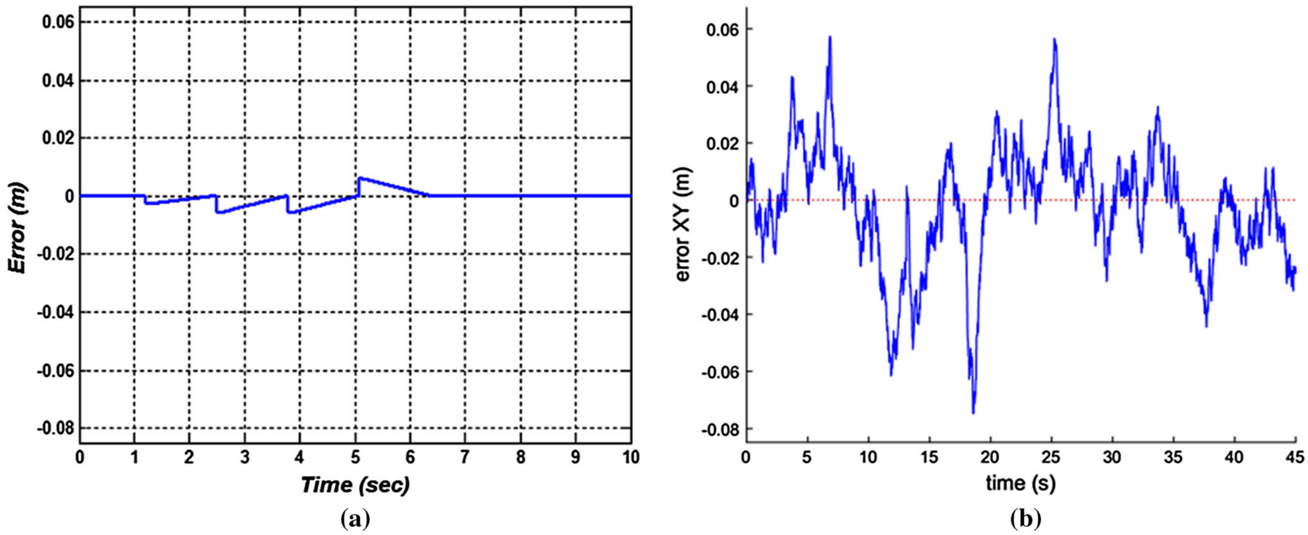


Fig. 25 The tracking error of the pulse trajectory: **a** the proposed approach and **b** the approach of Wang et al. (2018b)

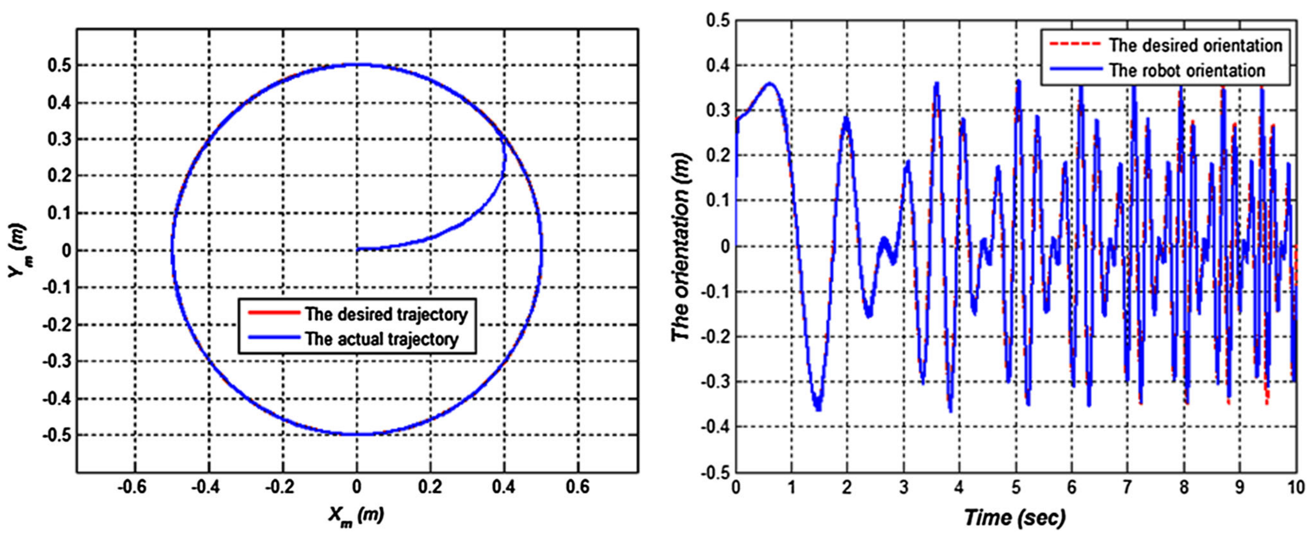


Fig. 26 Tracking a circular trajectory with oscillating orientation

5.4 Complex Scenarios

Figures 26, 27, 28 and 29 show different complex trajectories tracked successfully by the robot. In Fig. 26, the robot has to track a circular trajectory and track an oscillating orientation with a variable magnitude and variable time scale. In Fig. 27, the robot has to track a net-like trajectory while maintaining the orientation set to zero. In Fig. 28, the robot tracks a flower-like trajectory, while its orientation follows a sine trajectory. Figure 29 shows a complex trajectory in which the robot successfully tracks a dragonfly-like trajectory, while the orientation tracks an oscillating trajectory of varying magnitude.

6 Conclusion

A detailed mathematical analysis of the motion of a three-wheeled omnidirectional mobile robot was performed. The analysis led to the kinematics of the robot. The motion of the TWOMR can be divided into three types: a pure rotation

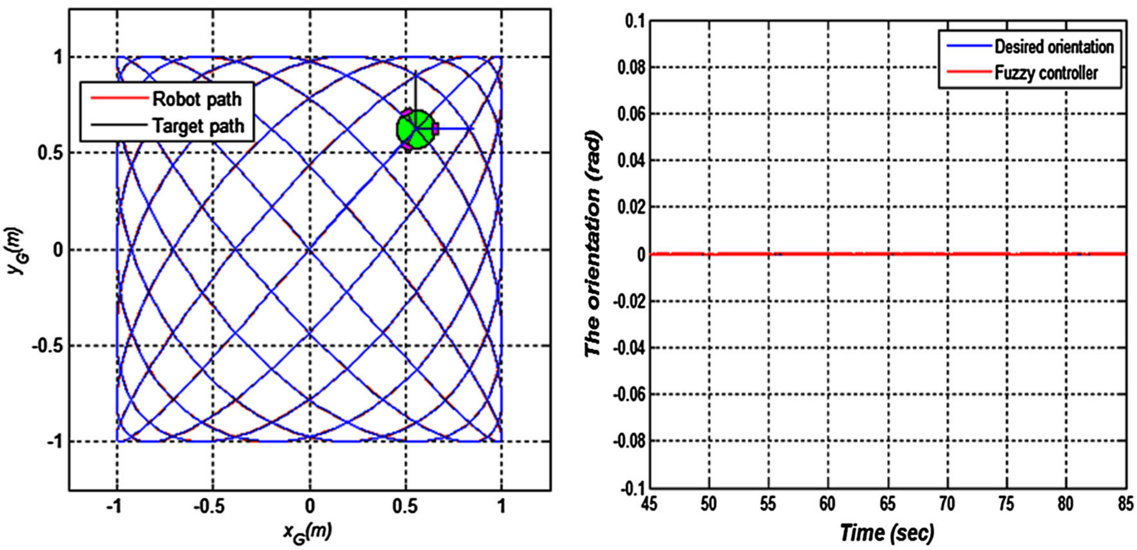


Fig. 27 Net-shaped trajectory with zero angle orientation

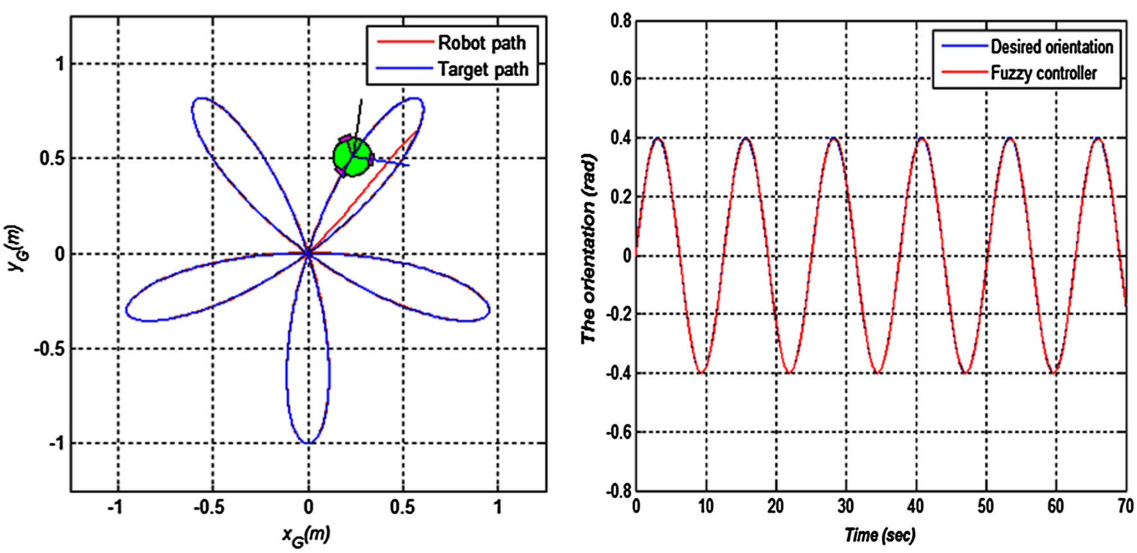


Fig. 28 Flower-shaped trajectory with sine orientation

around the robot center, a pure translational motion and a rotation around a point different from the robot center with a nonzero radius. A fuzzy controller has been designed to the trajectory tracking problem which allows the robot to track the desired trajectory, while its orientation also tracks the desired orientation. The effectiveness and flexibility of the approach proposed for the trajectory tracking problem of the TWOMR are verified through numerous tests. The comparison with another controller from the literature shows the advantage of the fuzzy controller with a minimum number of fuzzy rules (only four rules) compared to a model predictive controller, so the best solution is perhaps the simplest. In addition, all motor speeds are within the range of the allowable controls, which demonstrates the effectiveness of the proposed solution to the saturation problem. Future work is on multi-robot systems.

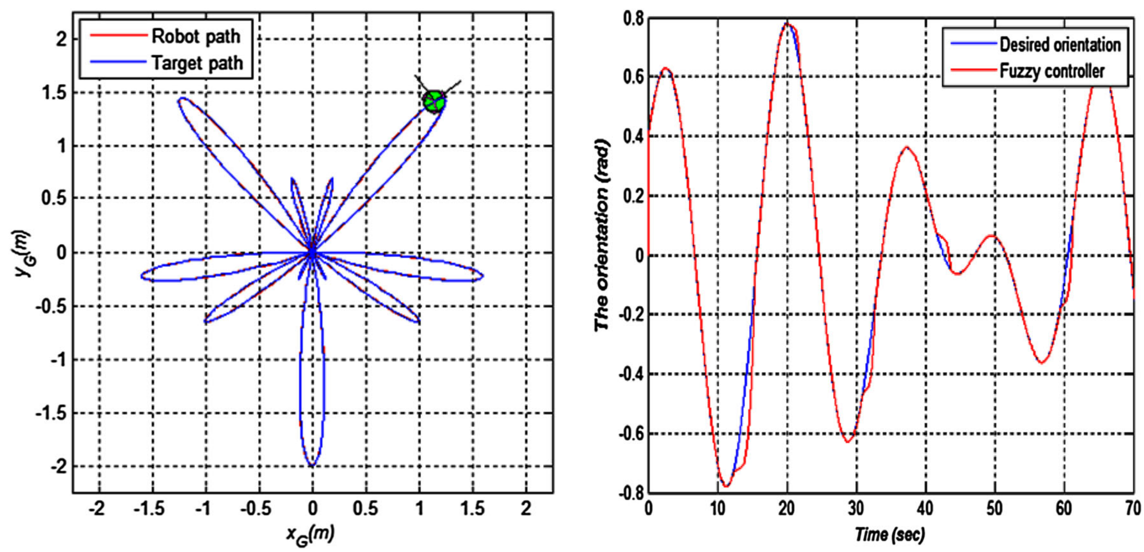


Fig. 29 Dragonfly-shaped trajectory with an oscillating orientation

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